

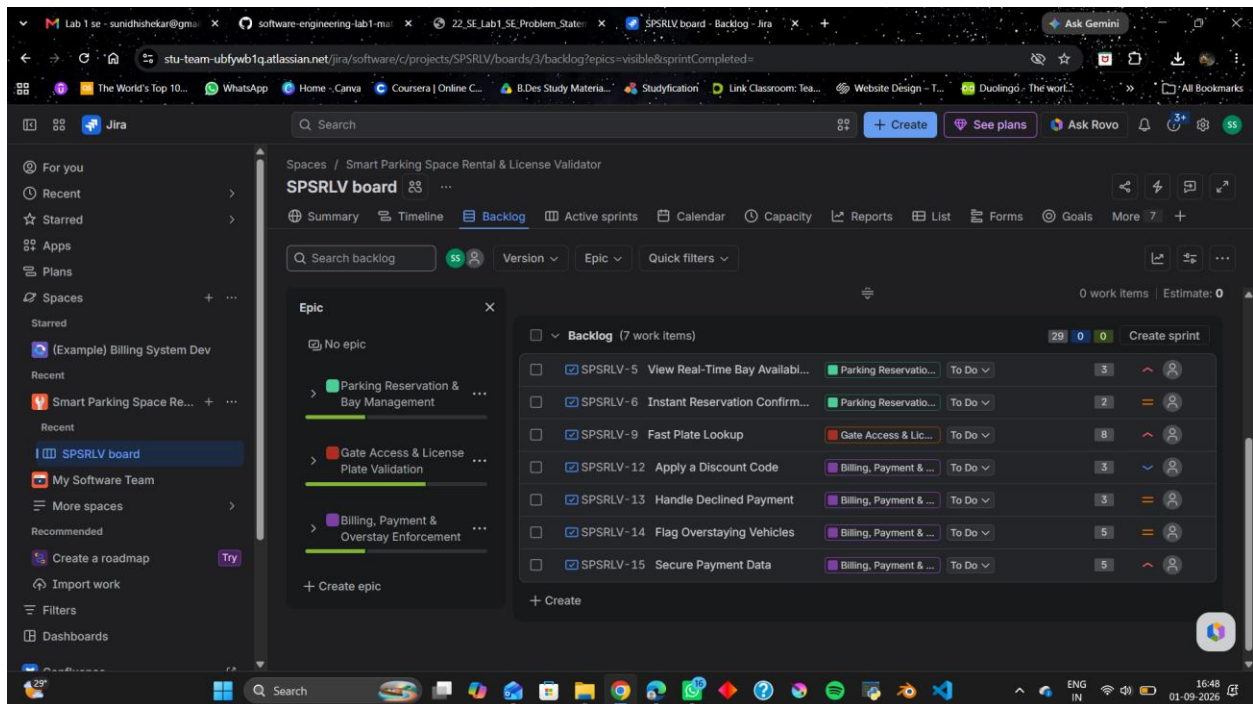
SE Lab 2: Agile Backlog Creation and Sprint Simulation in Jira

22. Smart Parking Space Rental and License Validator

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Backlog

The product backlog was structured into three Epics derived from Lab 1's Functional and Non-Functional Requirements: Parking Reservation and Bay Management, Gate Access and License Plate Validation, and Billing, Payment and Overstay Enforcement. Each Epic was broken into User Stories, prioritized, and estimated using story points.



Product backlog showing all Epics with their story points and progress bars.

Sprint 1

Sprint 1 was scoped to deliver the system's core end to end flow: reserving a bay, validating a vehicle at entry, calculating the parking fee, and completing payment at exit. Five User Stories, totalling 28 story points, were selected from the backlog and moved into Sprint 1.

Sprint Goal

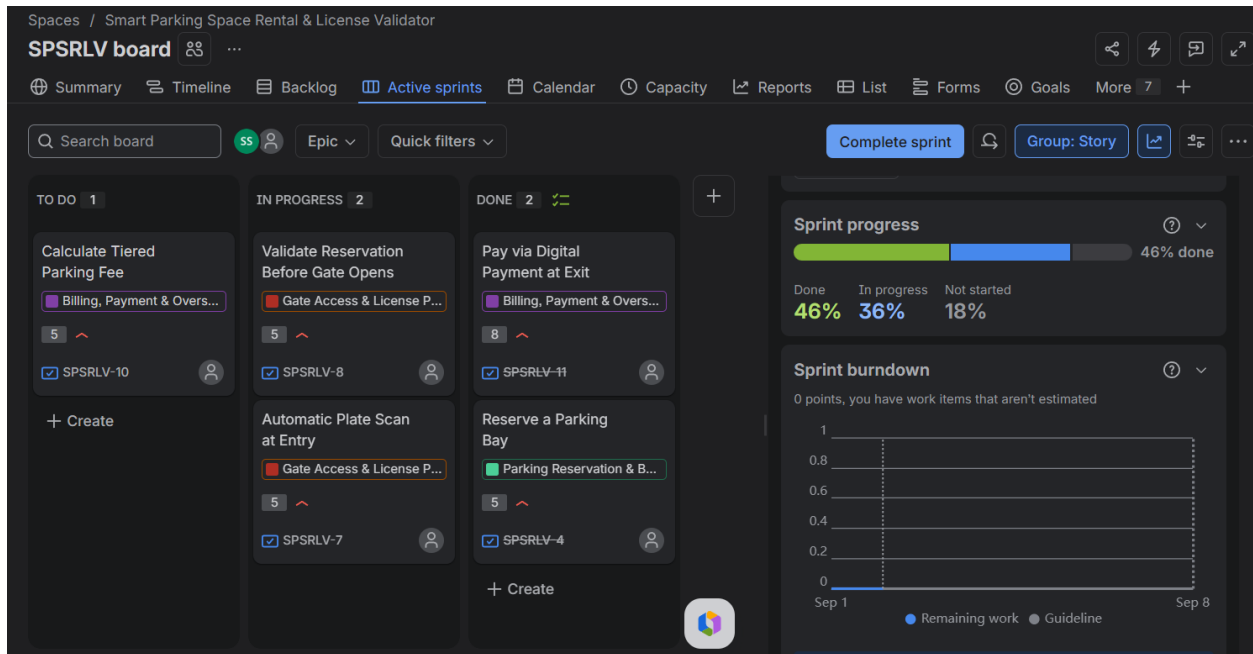
Deliver the core reservation to payment flow: bay reservation, gate entry validation, fee calculation, and digital payment.

- Reserve a Parking Bay, 5 points
- Automatic Plate Scan at Entry, 5 points
- Validate Reservation Before Gate Opens, 5 points
- Calculate Tiered Parking Fee, 5 points

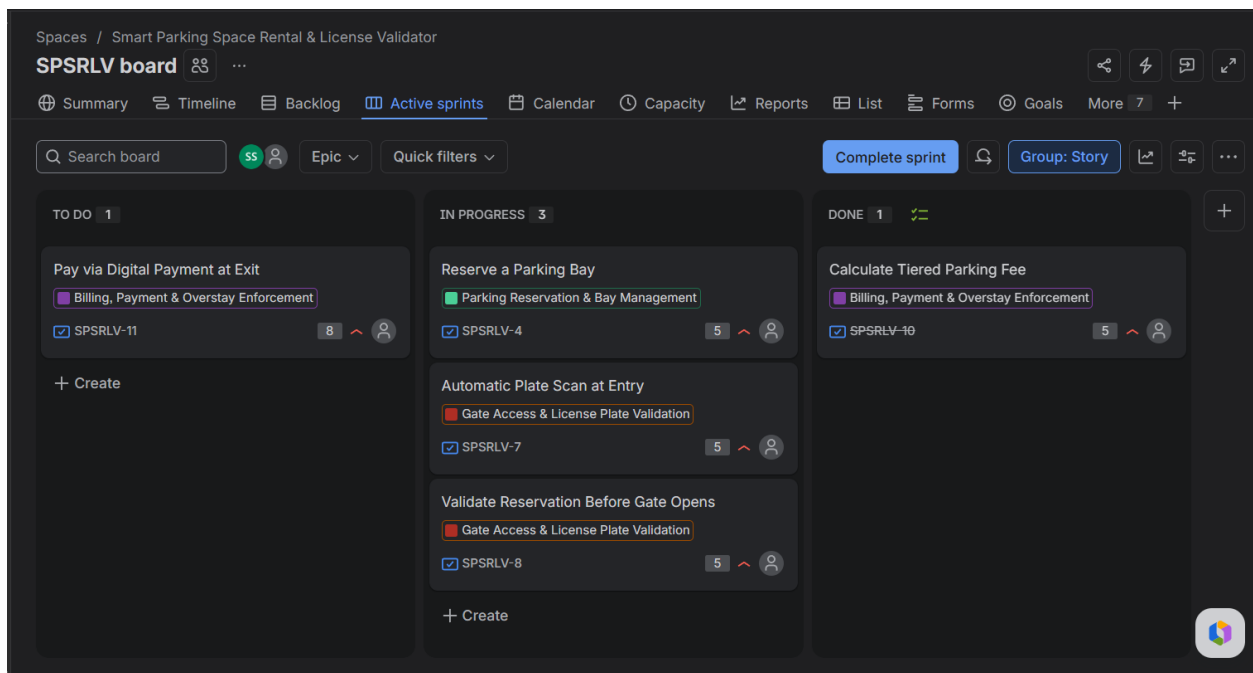
- Pay via Digital Payment at Exit, 8 points

Sprint 1 Progress

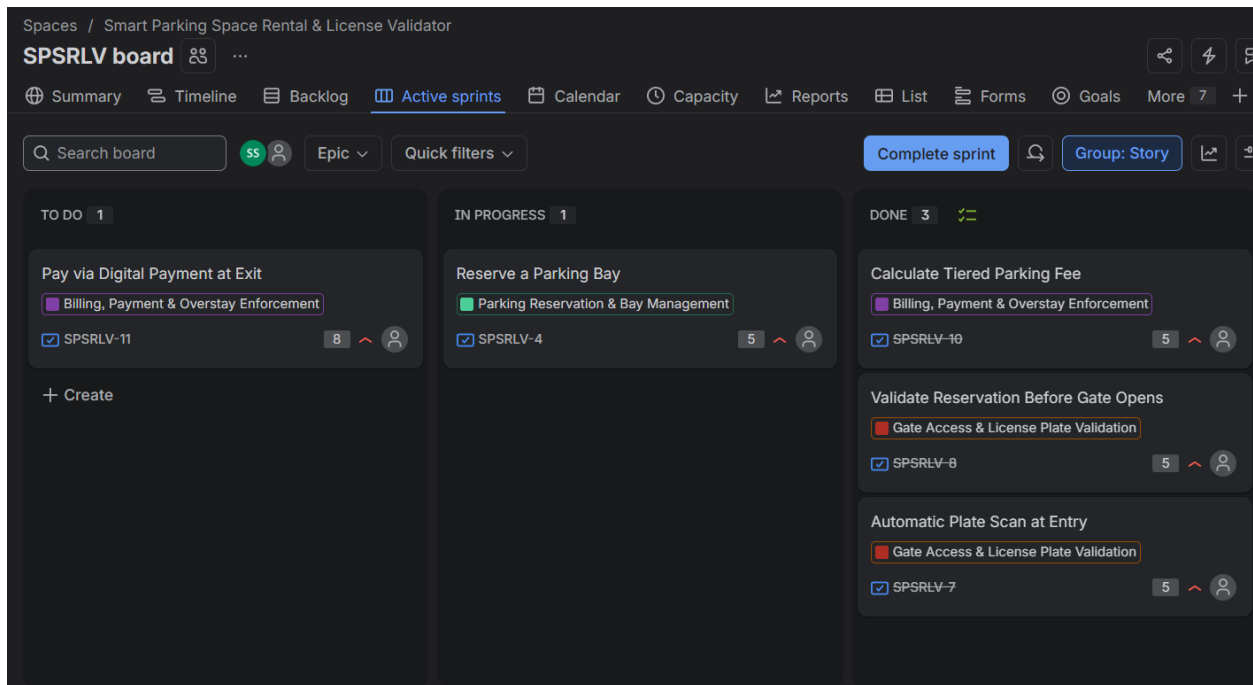
Stories were moved incrementally across the board, from To Do to In Progress to Done, to simulate real sprint execution. The following snapshots were captured along the way.



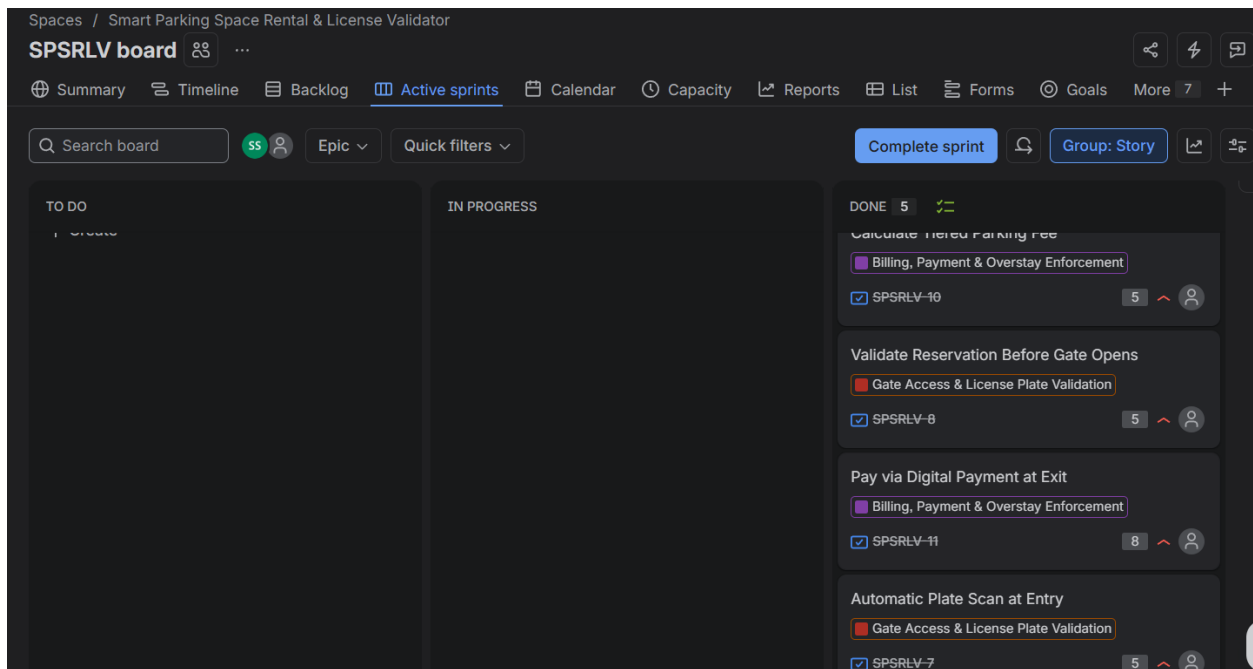
Sprint progress snapshot, showing 2 of 5 stories done, 2 in progress, 1 to do.



Later snapshot, showing 1 story done, 3 in progress, 1 to do.



Near end of sprint, showing 3 of 5 stories done, 1 in progress, 1 to do.

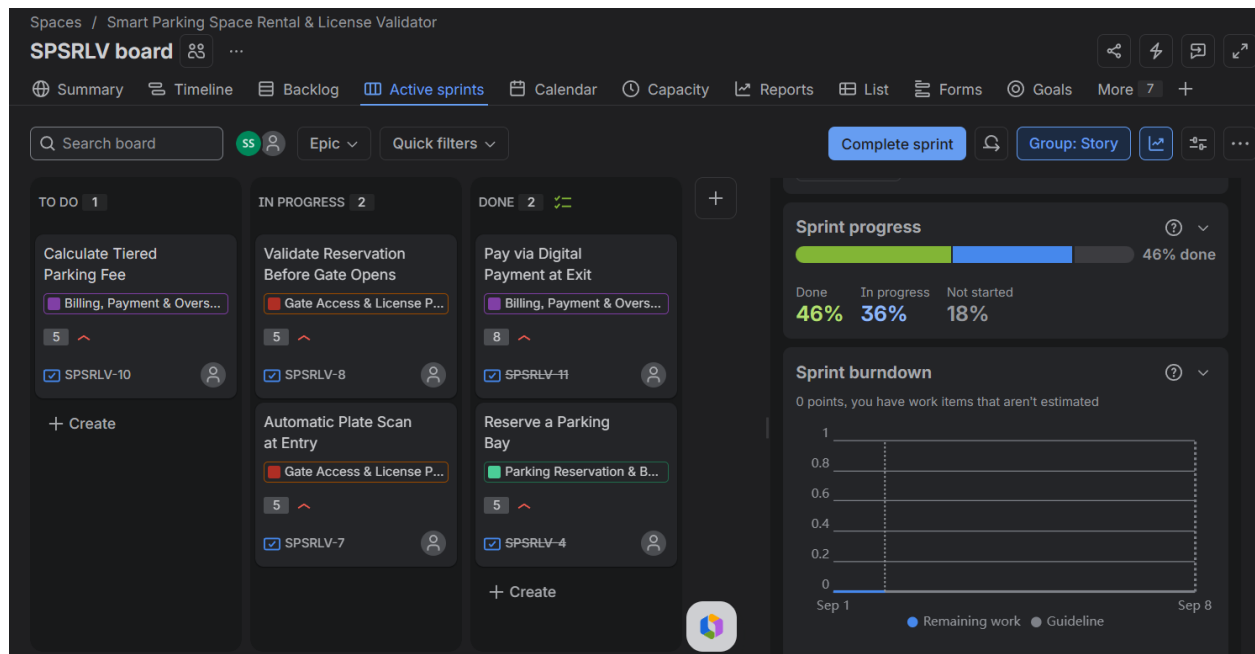


All 5 stories moved to done, sprint work complete.

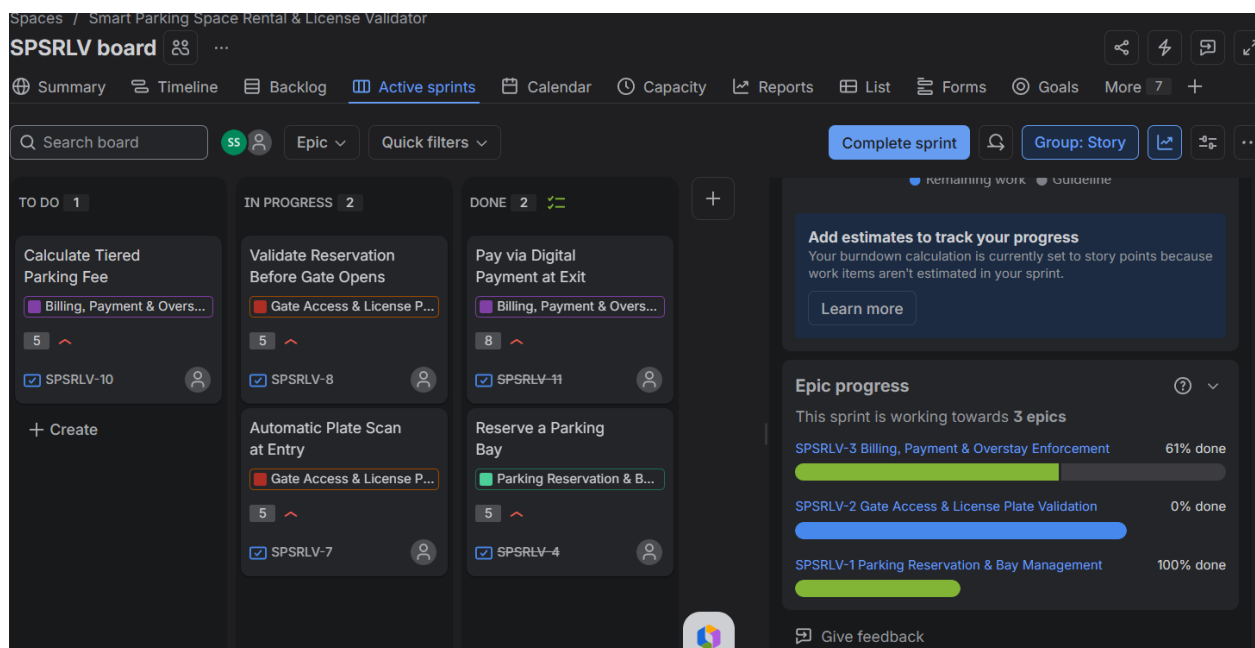
Sprint 1 Burndown

The Sprint Progress panel recorded 46 percent of work done, 36 percent in progress, and 18 percent not started at that point in the sprint. Because Jira's burndown chart updates through daily snapshots rather than in real time, the burndown line for a compressed simulated sprint

does not visibly step down until the sprint is completed. This is expected behavior, not a data error.



Sprint 1 burndown chart, showing remaining work against the guideline line.



Sprint 1 burndown chart with the Epic Progress panel, showing Parking Reservation and Bay Management at 100 percent done.

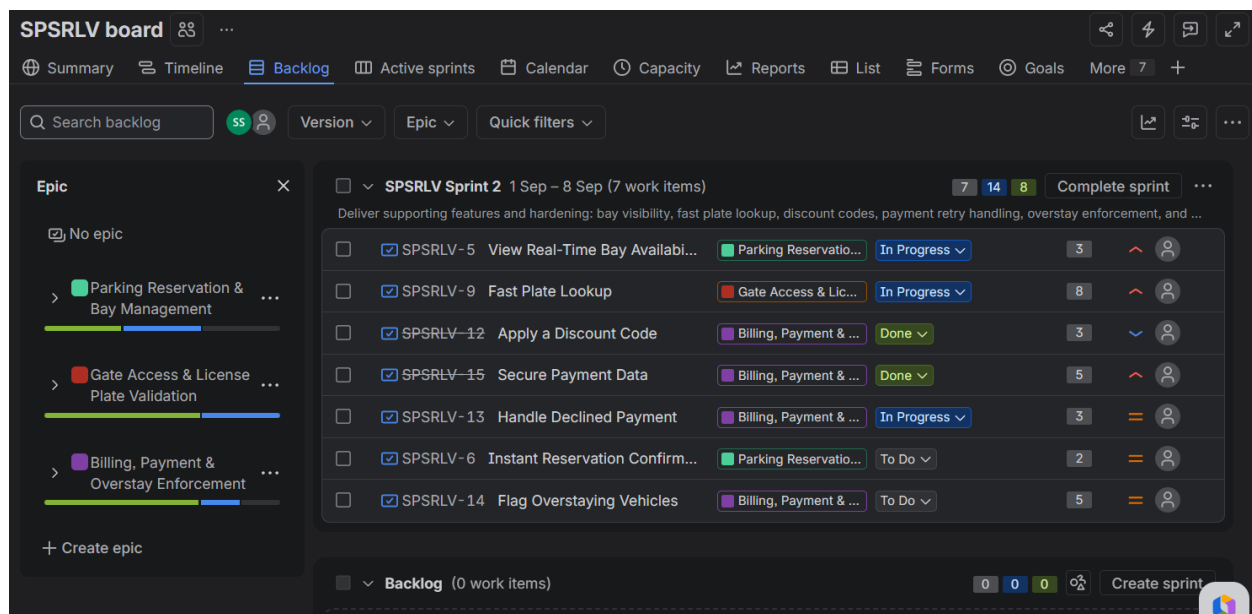
Sprint 2

Sprint 2 picked up the remaining seven User Stories from the backlog, covering supporting features and hardening work that builds on the core flow delivered in Sprint 1. Total scope was 29 story points.

Sprint Goal

Deliver supporting features and hardening: bay visibility, fast plate lookup, discount codes, payment retry handling, overstay enforcement, and payment data security.

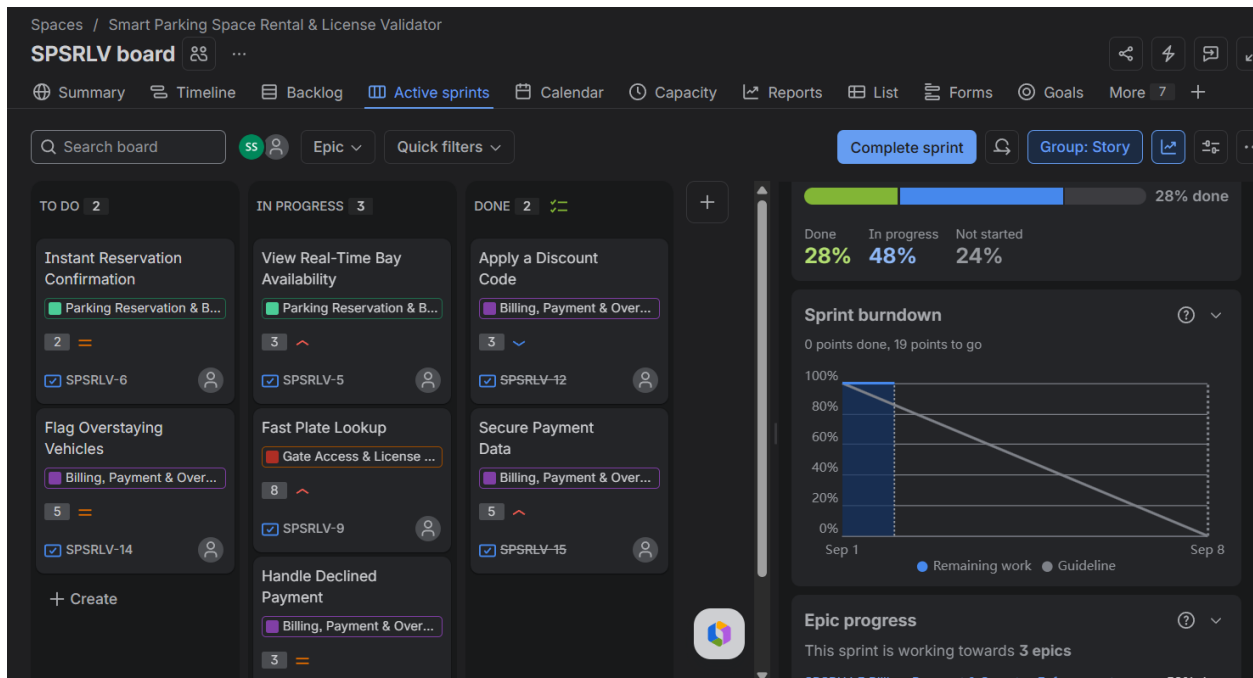
- View Real Time Bay Availability, 3 points
- Instant Reservation Confirmation, 2 points
- Fast Plate Lookup, 8 points
- Apply a Discount Code, 3 points
- Handle Declined Payment, 3 points
- Flag Overstaying Vehicles, 5 points
- Secure Payment Data, 5 points



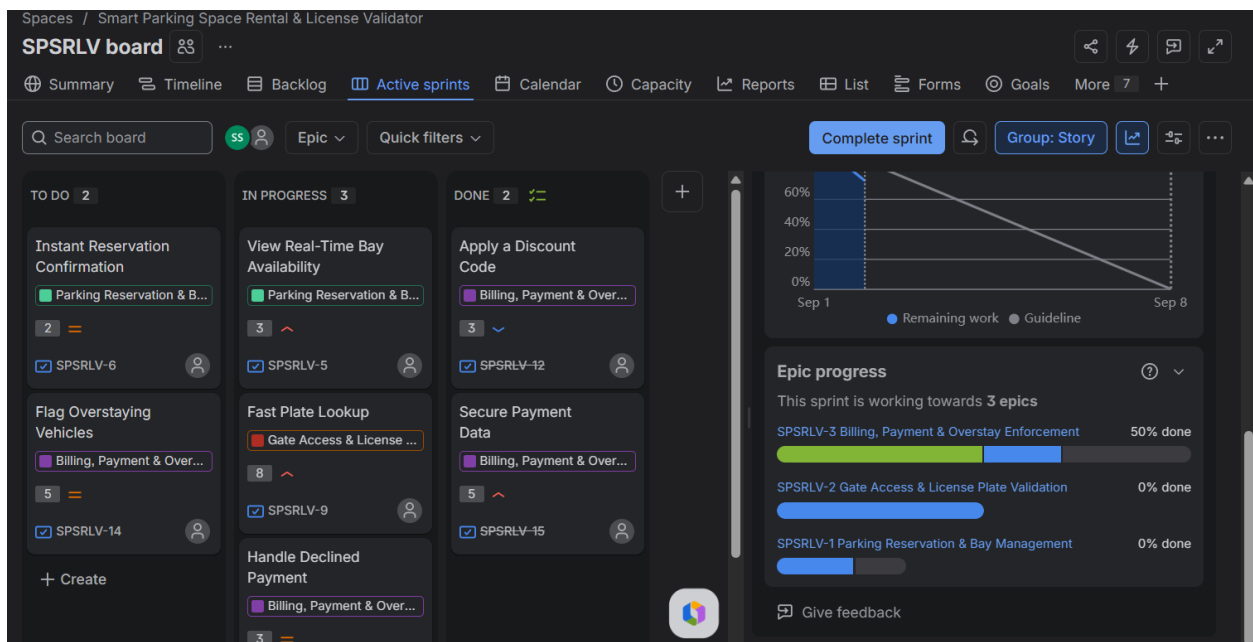
Sprint 2 backlog list view, showing all seven stories with story points and starting statuses.

Sprint 2 Progress

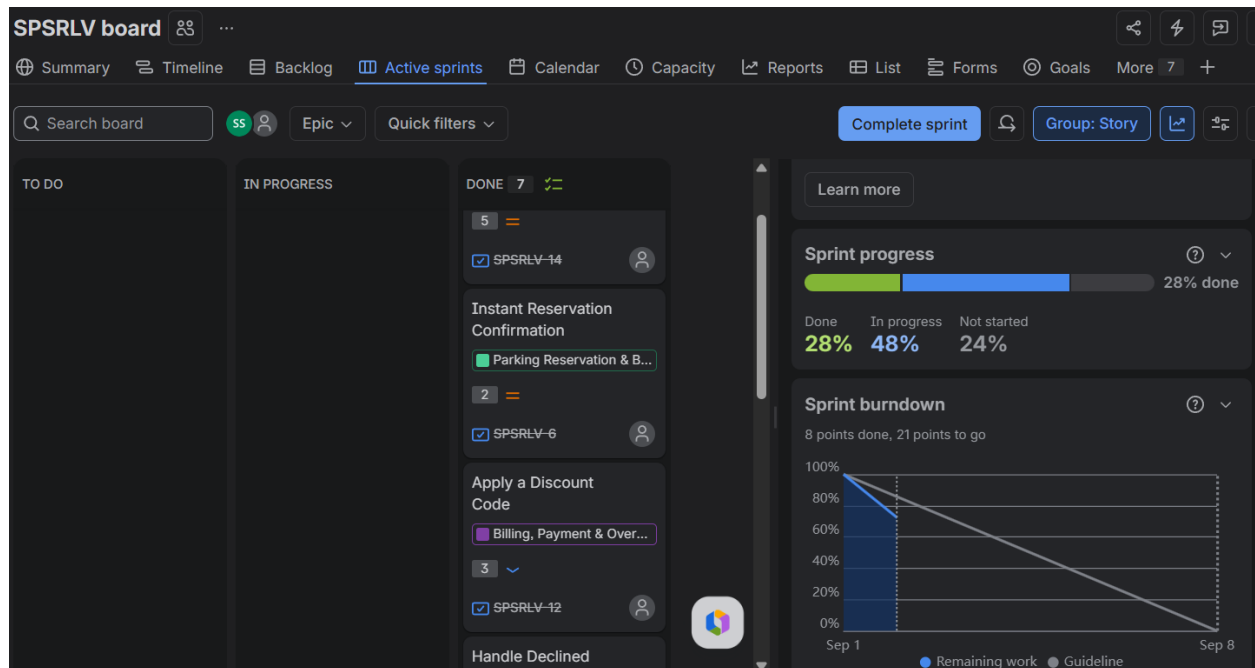
Stories were moved incrementally across the board to simulate execution, following the same approach as Sprint 1.



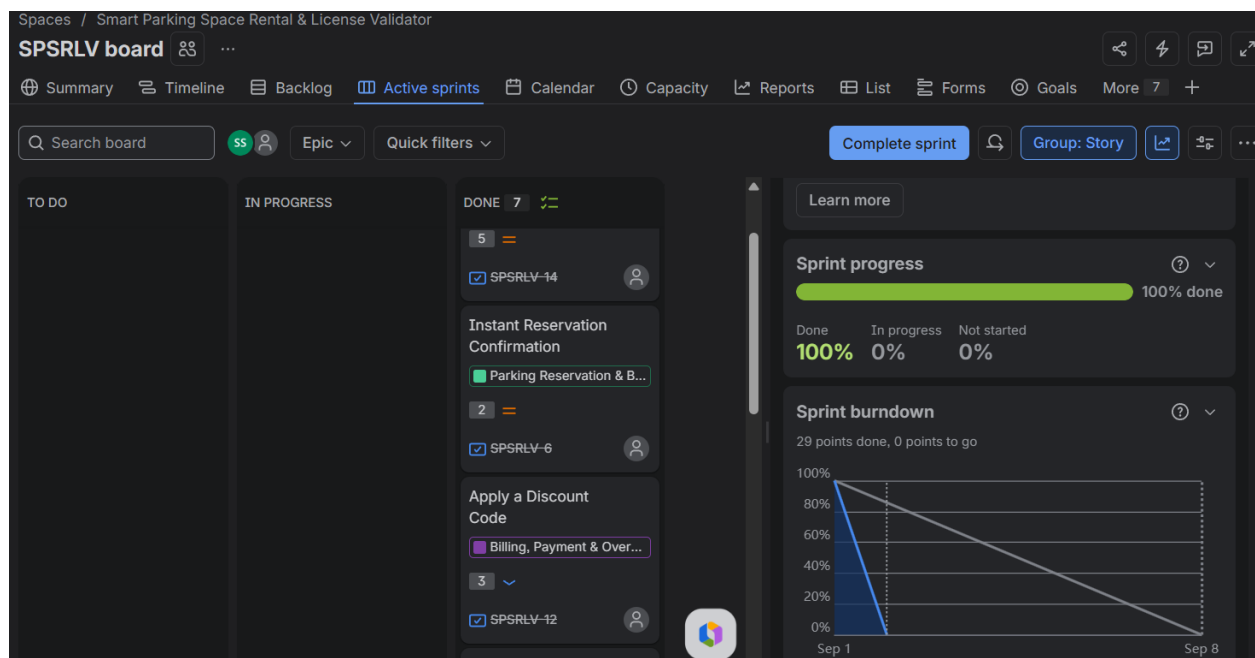
Early sprint snapshot, showing 2 stories done, 3 in progress, 2 to do.



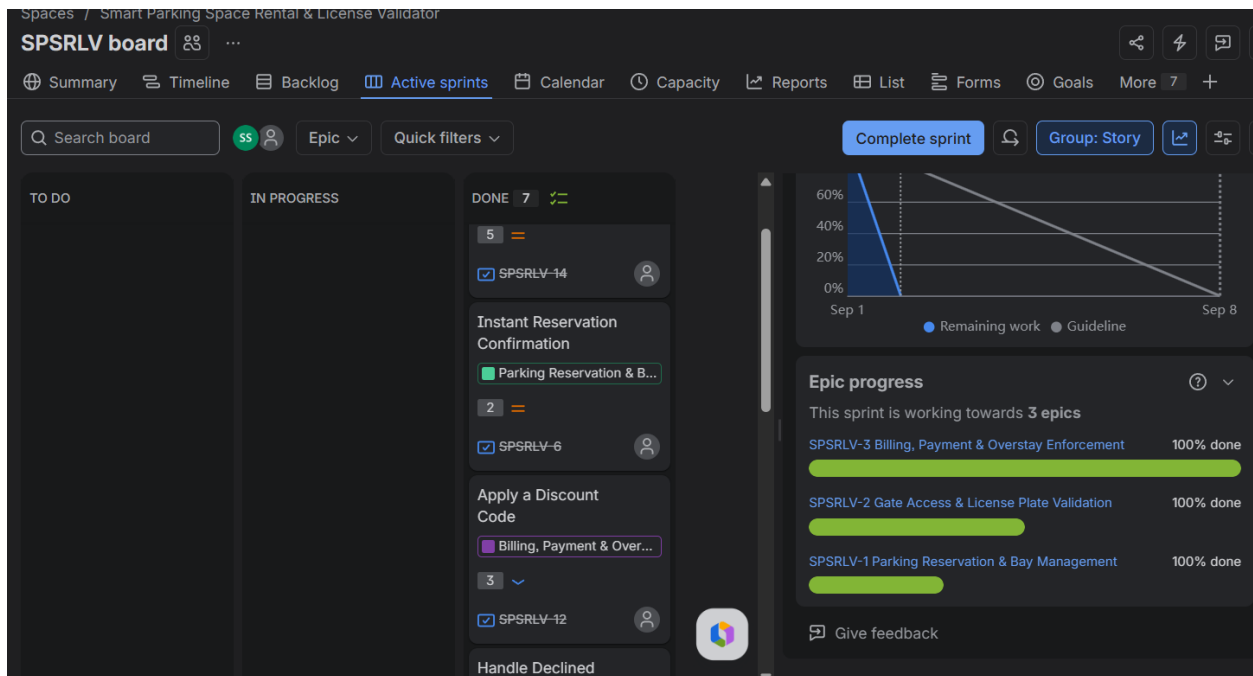
Same snapshot, showing the Epic Progress panel with the Billing, Payment and Overstay Enforcement Epic at 50 percent done.



Later snapshot, showing all 7 stories moved into the done column.



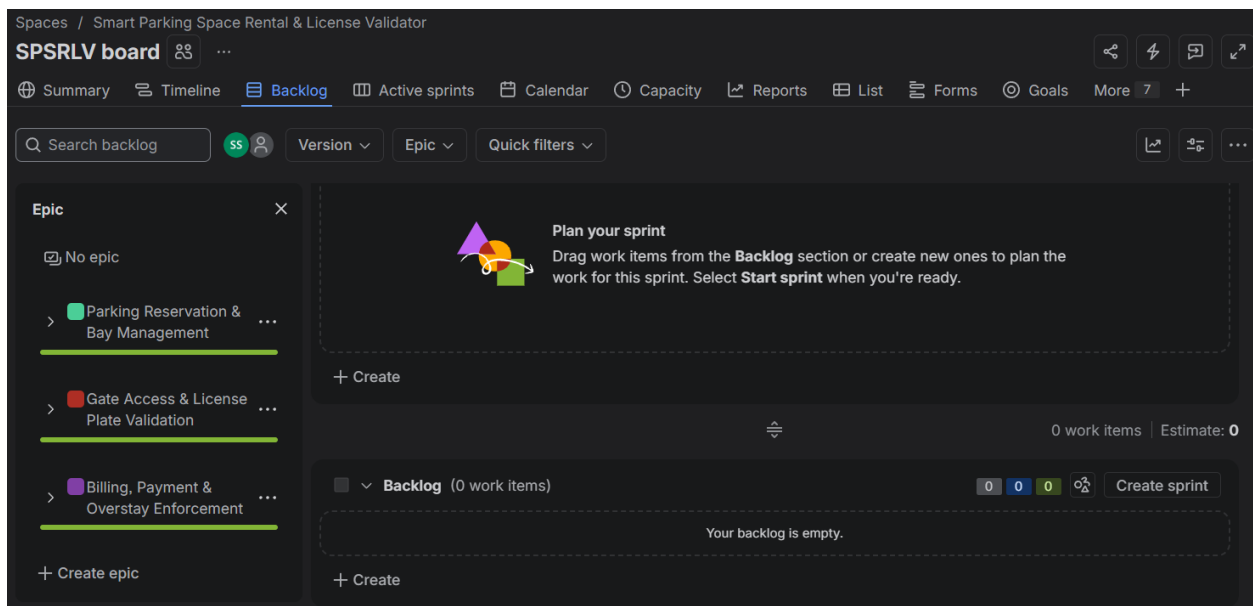
Sprint Progress panel updated, showing 100 percent done and all 29 points completed.



Epic Progress panel confirming all three Epics at 100 percent done.

Sprint 2 Completion

With all 7 stories in done and every Epic showing 100 percent completion, Sprint 2 was closed. The backlog view afterward confirms zero remaining work items and all three Epics fully complete.



Backlog view after completing both sprints, showing full progress bars across all Epics and an empty backlog.

Reflection

Did your estimations reflect the actual effort

Yes. Higher uncertainty stories, such as Fast Plate Lookup and Pay via Digital Payment at Exit, both at 8 points, were the largest items in each sprint, consistent with their integration complexity. Smaller, well understood stories like Instant Reservation Confirmation and Apply a Discount Code were estimated and completed with comparatively little friction.

Was your backlog well prioritized

Yes. Sprint 1 grouped the five high priority stories that form one coherent core flow: reserve, enter, validate, calculate fee, and pay. Sprint 2 deferred medium and low priority supporting and hardening stories, such as discounts, overstay flagging, payment retry, and encryption, which add robustness but are not required for a minimal usable flow.

How did your simulated sprint align with your plan

Both sprints closed with all planned scope completed: 28 of 28 points in Sprint 1 and 29 of 29 points in Sprint 2. Every story moved fully through To Do, In Progress, and Done, and every associated Epic reached 100 percent completion. The simulated sprints matched the planned scope exactly, with no stories removed, added mid sprint, or carried over.

What insights did the burndown chart give about your team's capacity

Because both sprints were simulated in minutes rather than real days, the burndown line stayed flat for most of the window and only dropped near the end. This is a known limitation of Jira's daily snapshot based burndown tracking, not a reflection of real team velocity. In an actual sprint run over calendar days, the same total effort would burn down more gradually, with the remaining work line tracking closer to the guideline line throughout the week.